

ADHEESH TRIVEDI

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MTECH ARTIFICIAL INTELLIGENCE '28, IISc

EDUCATION

IISER Bhopal | GPA: 9.34/10 (3.74/4) 2022 – 2026
B.S. | Computer Science & Engineering Bhopal, India

Discrete Mathematics	Signals & Systems	Modern Cryptography
Probability & Statistics	Machine Learning	Information Theory & Coding
Complex Variables	Computer Vision	Computer Organization
Group Theory	Artificial Intelligence	Network Sciences
Real Analysis	Database Systems	Operating Systems
Linear Algebra	Introduction to C Programming	Computer Networks
Multivariable Calculus	Data Structures & Algorithms	Advanced Programming
Applied Optimization	Algorithms (Advanced)	Compiler Design
Theory of Computation	Software Modelling & Verification	

Indian Institute of Science | GPA: dnf* 2026 – 2028
M.Tech. | Artificial Intelligence Bangalore, India

TECHNICAL PROJECTS

-  **Nifty Network Analysis & Visualization Dashboard** Oct 2025 – Nov 2025
 - Analyzed relationships among major Nifty companies and summarized insights in an interactive dashboard and formal report.
 - Delivered clear visual findings on market structure, influential firms, and cluster patterns for course project.
-  **Extensively Customizable Exam Scheduler using Graph Coloring** Oct 2024 – Jan 2025

Vivek Kumar, Rahul Jana, Ayushman Shaha, [Dr. Prafullkumar Tale](#)

 - Developed an engine that assigns exams to time slots & halls minimizing same-day conflicts ($\leq 24h$).
 - Uses graph coloring and randomized optimization heuristics for reaching close to optimal scheduling.
 - Supports multi-hall allocation when enrollment exceeds single capacity.
-  **Texture Classification & Face Clustering for Image Search** Sep 2024 – Nov 2024
 - Addresses the common challenge of navigating through a large collection of images,
 - Enables the user to efficiently filter and search for images by face or texture.
 - Implemented batching to control GPU memory and multiple texture descriptors for retrieval precision.
-  **GUI Developer Intern** Feb 2024 – Aug 2024

Mentor: [Prof. Vardharajan Srinivasan](#) AITG Labs

 - Led integration of DFT engines ([Octopus](#), [GPAW](#), [NWChem](#)) within a Python toolkit for photo-induced phenomena simulations.
 - Automated submission workflows and job orchestration on HPC clusters.
-  **Context aware Bag-of-Words Chatbot** Oct 2021 – Dec 2021
 - Built a context-aware chatbot supporting tasks like to-do lists, word definitions, note taking, make jokes and time queries across timezones.
 - Trained a sequential neural network in TensorFlow on a custom dataset tailored to project requirements.
-  **General Purpose Discord Bot** Mar 2021
 - Python project that utilized PostgreSQL to implement feature-rich Discord bot.

*Ongoing

RESEARCH PROJECTS

-  **Practical study on embeddings between KS and LTS** Dec 2025 – Apr 2026
Mentor: [Dr. Arpit Sharma](#), PhD. Shonak Saha
- Built a Python framework for model checking of explicit LABELLED TRANSITION SYSTEMS models
 - Supported multiple backends (NuSMV, Spin, Storm) by embedding into KRIPKE STRUCTURES.
 - Benchmarked and studied the feasibility of explicit model checking in embedded models.
-  **On Robust Coloring of Graphs and ETH** July 2025 – Jan 2026
Mentor: [Dr. Prafullkumar Tale](#)
- Studied a relaxation of proper coloring in which edges with same color are associated with some cost.
 - Studied the EXPONENTIAL TIME HYPOTHESIS, and ETH-Preserving reductions.
 - Investigated treewidth; applied tree decompositions to dynamic programming on hard problems.
-  **Reading Project on Graph Theory** Jan 2024 – Apr 2024
Mentor: [Dr. Prafullkumar Tale](#)
- Studied *A First Look at Graph Theory* (Clark & Holton) with problem solving.
 - Reinforced concepts: Graphs & their types, connectivity, traversals, matching problem, planarity.

SKILLS

Python • C++ (CMake, GoogleTest) • Git & GitHub • L^AT_EX • typst • Linux Shell • CI/CD • React • Rust

TALKS & TEACHING

- Teaching Assistant - Theory of Computation** Aug 2025 – Dec 2025
Instructed by [Dr. Arpit Sharma](#)
-  **Fast Matrix Multiplication Algorithms** 1 Sept 2023
Math Club @ IISER Bhopal

ACADEMIC ACHIEVEMENTS

- Secured **rank 29** (99.96 %ile) in GATE Data Science & Artificial Intelligence 2026
- Secured **rank 147** (99.93 %ile) in GATE Computer Science 2026
- Scored **92%** in 10th board (ICSE) and **91.2%** in 12th board (CBSE)

LEADERSHIP & ENGAGEMENT

- Club Coordinator** May 2024 – May 2025
Coding club @ IISER Bhopal
- Hackathon Organizer -- Armacode 0** Jan 2024 – Apr 2024
IISER Bhopal x IIIT Bhopal

CO-CURRICULAR

Music (Guitar) • Competitive Programming • Speed Typing